

Deliverable 8 – DAX Development Report

Healthcare Patient Experience Analytics using Power BI

Introduction

Data Analysis Expressions (DAX) is the formula language used in Microsoft Power BI to create calculated measures and perform advanced analytical calculations. In this project, DAX measures were developed to evaluate healthcare performance, calculate patient satisfaction KPIs, perform benchmarking, rank states and regions, and support interactive dashboard reporting.

The DAX measures enable healthcare stakeholders to analyze patient experience across multiple dimensions, compare healthcare performance over time, and make evidence-based decisions through dynamic visualizations.

Objectives of DAX Development

The primary objectives of developing DAX measures in this project were to:

- Calculate healthcare performance indicators.
 - Compare national and state-level patient satisfaction.
 - Rank states and regions based on healthcare performance.
 - Measure year-over-year improvements.
 - Perform variance analysis against national benchmarks.
 - Enable dynamic dashboard interactions.
 - Support executive reporting with meaningful KPIs.
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DAX Measures Developed

1. Average Top-box %

Definition

Calculates the average percentage of patients selecting the most positive survey response (Top-box).

Purpose

Measures overall patient satisfaction and serves as a key healthcare performance indicator.

2. Average Middle-box %

Definition

Calculates the average percentage of patients selecting the Middle-box response category.

Purpose

Represents moderate patient satisfaction and helps identify opportunities for service improvement.

3. Average Bottom-box %

Definition

Calculates the average percentage of patients selecting the least positive survey response.

Purpose

Highlights patient dissatisfaction and identifies healthcare areas requiring immediate attention.

4. National Average

Definition

Calculates the average patient satisfaction across all participating states.

Purpose

Provides a national benchmark for comparing healthcare performance.

5. State Average

Definition

Calculates the average patient satisfaction score for the selected state.

Purpose

Measures healthcare performance at the state level and supports benchmarking.

6. Satisfaction Index

Definition

Calculates an overall patient satisfaction score based on survey responses.

Purpose

Provides a consolidated performance indicator for healthcare quality evaluation.

7. Response Rate

Definition

Calculates the percentage of survey responses received.

Purpose

Measures participation in the HCAHPS survey and indicates the reliability of survey results.

8. State Rank

Definition

Ranks participating states based on patient satisfaction scores.

Purpose

Identifies high-performing and low-performing states.

9. Region Rank

Definition

Ranks healthcare regions according to average patient satisfaction.

Purpose

Supports regional benchmarking and comparative analysis.

10. Year-over-Year Change

Definition

Calculates the change in patient satisfaction between reporting periods.

Purpose

Measures long-term healthcare performance improvement.

11. Dynamic Rankings

Definition

Updates state and region rankings dynamically based on user selections.

Purpose

Provides interactive ranking analysis using Power BI slicers and filters.

12. Variance from National Average

Definition

Calculates the difference between state performance and the national average.

Purpose

Highlights states performing above or below the national benchmark.

13. Dynamic Titles

Definition

Generates report titles that automatically update according to the selected filters.

Purpose

Improves dashboard usability and provides context-aware reporting.

Business Benefits of DAX Measures

The developed DAX measures provide the following business benefits:

- Automate KPI calculations.
 - Enable real-time performance monitoring.
 - Improve dashboard interactivity.
 - Support national and regional benchmarking.
 - Simplify executive reporting.
 - Improve analytical accuracy.
 - Enable dynamic filtering and drill-down analysis.
 - Support evidence-based healthcare decision-making.
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DAX Development Best Practices

The following best practices were followed while developing DAX measures:

- Created reusable measures instead of calculated columns wherever possible.
 - Used meaningful measure names for readability.
 - Organized measures into logical groups.
 - Optimized calculations to improve report performance.
 - Avoided unnecessary nested calculations.
 - Ensured consistency across all dashboard reports.
 - Designed measures to respond dynamically to slicers and filters.
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Summary of Developed Measures

Category	Measures
National KPIs	Average Top-box %, Average Middle-box %, Average Bottom-box %, National Average
State KPIs	State Average, State Rank
Regional KPIs	Region Rank
Performance KPIs	Satisfaction Index, Response Rate
Trend KPIs	Year-over-Year Change
Benchmark KPIs	Variance from National Average
Dynamic Features	Dynamic Rankings, Dynamic Titles

Conclusion

The DAX measures developed for this project provide the analytical foundation for the Healthcare Patient Experience Analytics dashboard. These measures transform raw healthcare survey data into meaningful KPIs that enable executives, healthcare administrators, and policy makers to monitor patient satisfaction, benchmark healthcare performance, analyze trends, and identify opportunities for continuous improvement. The combination of optimized DAX calculations and interactive Power BI visualizations ensures accurate, scalable, and efficient business reporting.

